

In-Class Exercise/Demo



In-Class Demo

- Let's practice more React.
- First, create a new Github repository **m5-w1-d2-demo** and clone this in the WESTCLIFF folder.
- Head over to GAP, download [m5-w1-d2.zip](#), unzip and drop the content into the clone folder.
- Launch VS Code and navigate to the clone folder.
- Open [demo1.html](#).
- You will notice we added React, React-DOM and Babel javascript CDN links at the bottom of the page.

```
<script src="https://unpkg.com/react@17/umd/react.development.js"></script>
<script src="https://unpkg.com/react-dom@17/umd/react-dom.development.js"></script>
<script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
```



In-Class Demo

- Open `demo1.js`
- We are going to create a `Car` class.
- There is `constructor`, and `super` props being used as well.
- We will be creating `multiple state variables` which we will use later on.
- Let's add the `state variables` below

```
this.state = {  
  brand: "Ford",  
  model: "Mustang ",  
  color: "red ",  
  year: 1964  
}
```



In-Class Demo

- Now let's render these variables using `this.state`

```
<div>
  <h1>My {this.state.brand}</h1>
  <p>
    It is a {this.state.color}
    {this.state.model}
    from {this.state.year}.
  </p>
</div>
```



In-Class Demo

- Open the [demo1.html](#) on live server.

My Ford

It is a red Mustang from 1964.



In-Class Demo

- Open [demo2.html](#).
- You will notice we added React, React-DOM and Babel javascript CDN links at the bottom of the page.
- Now let's open [demo2.js](#)
- Let's create a [changeColor](#) function after the constructor definition

```
changeColor = () => {  
  this.setState({color: "blue "});  
}
```



In-Class Demo

- Now we are going to create a button after the `</p>` tag that will update the color state to a value of blue. This action will be done without reloading the page.

```
<button  
  type="button"  
  onClick={this.changeColor}  
>Change color</button>
```



In-Class Demo

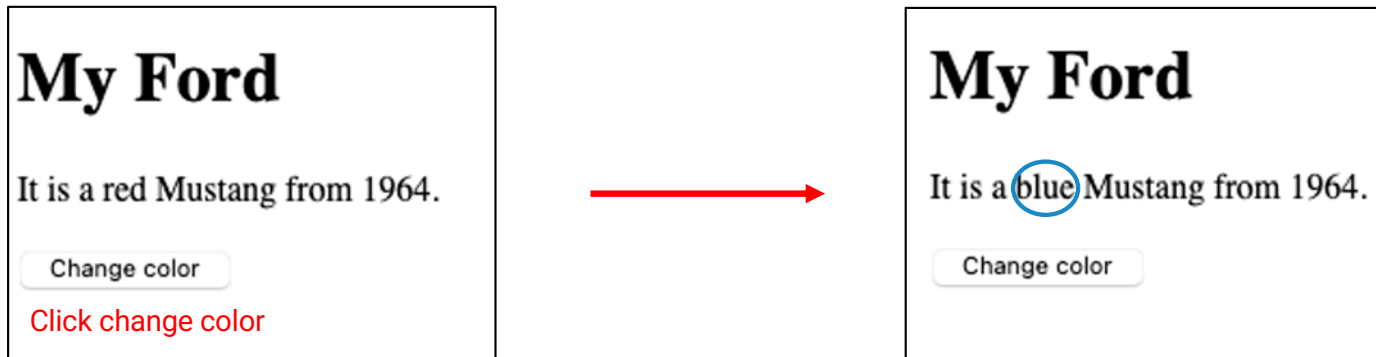
- Complete render code will look like this:

```
render() {  
  return (  
    <div>  
      <h1>My {this.state.brand}</h1>  
      <p>  
        It is a {this.state.color}  
        {this.state.model}  
        from {this.state.year}.  
      </p>  
      <button  
        type="button"  
        onClick={this.changeColor}  
      >Change color</button>  
    </div>  
  );  
}
```




In-Class Demo

- Open the page in live server





In-Class Demo

- Open [demo3.js](#).
- We are going to write a function, a [functional component](#) that will add an event

```
function clickbutton() {  
  alert("Clicked!!!!!!");  
}  
const myelement = (  
  <button onClick={clickbutton}>Click the button!</button>  
) ;
```

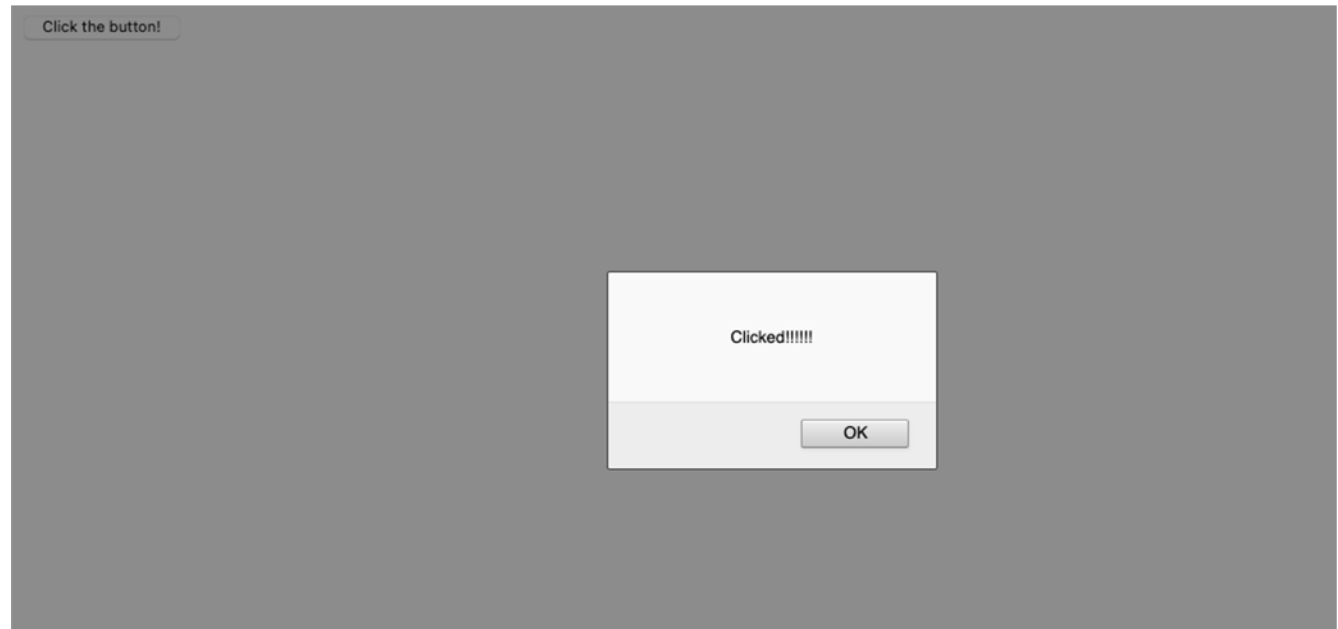
We are writing an **onClick** event that will run a **clickbutton** function once clicked. The function is firing an alert.



In-Class Demo

- Open the page in live server

Click the button!





In-Class Demo

- Open [demo4.js](#).
- We are going to convert our [functional component](#) to a [Class component](#).
- The [class Football](#) is rendering the button. Once [clicked](#), it executes the [shoot](#) function.

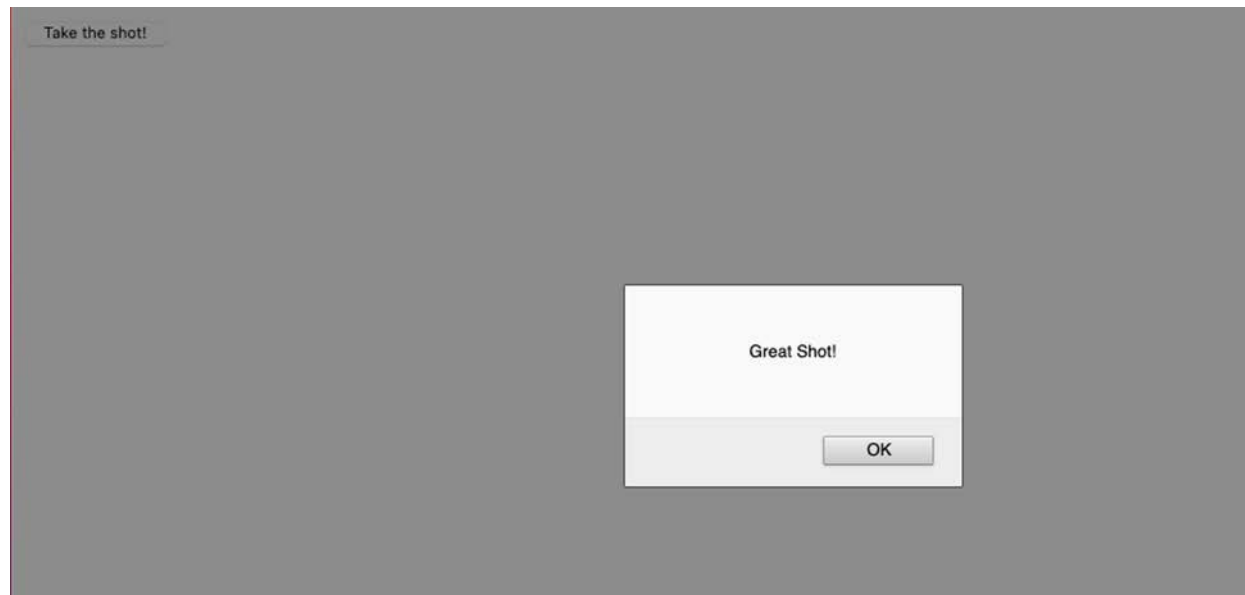
```
class Football extends React.Component {  
  shoot() {  
    alert("Great Shot!");  
  }  
  render() {  
    return (  
      <button onClick={this.shoot}>Take the shot!</button>  
    );  
  }  
}
```



In-Class Demo

- Open the page in live server

Take the shot!





In-Class Demo

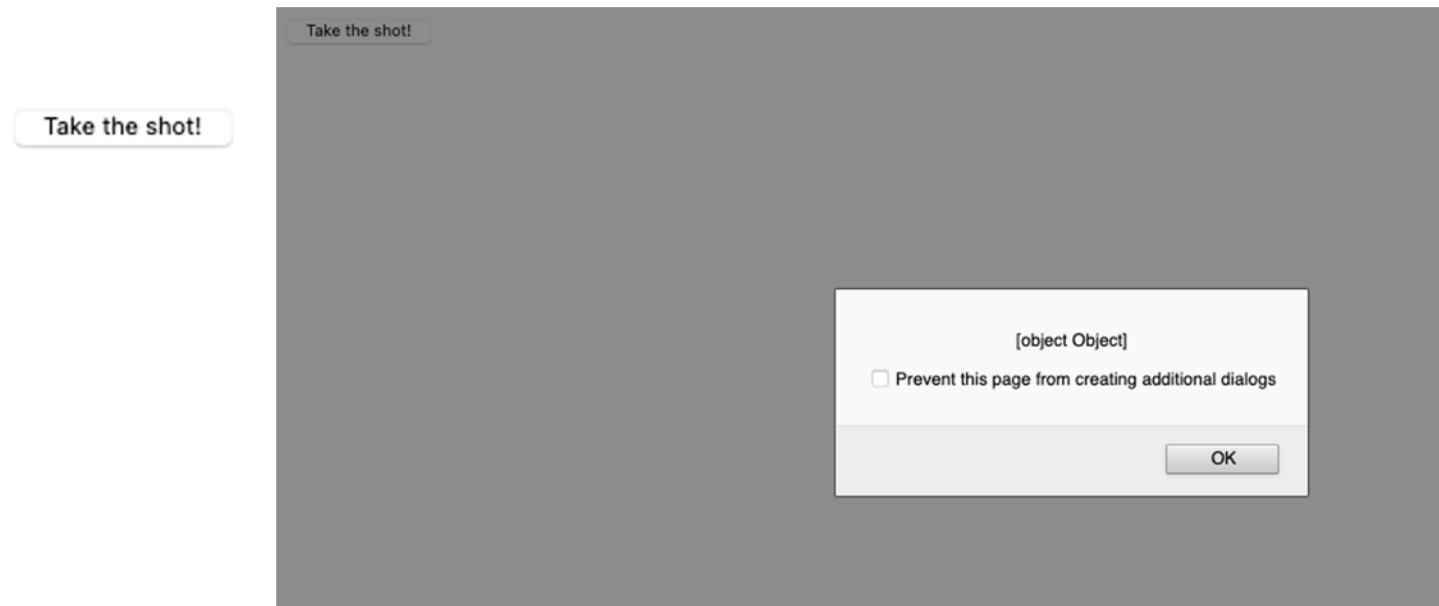
- Open [demo5.js](#).
- We are going to convert component to a [this](#) keyword with instance
- The [class Football](#) renders the button. Once [clicked](#), it executes the [shoot](#) arrow function.

```
class Football extends React.Component {
  shoot = () => {
    alert(this);
    /*
     The 'this' keyword refers to the component object
     */
  }
  render() {
    return (
      <button onClick={this.shoot}>Take the shot!</button>
    );
  }
}
```



In-Class Demo

- Since, we never set what `this` is going to be. It will just print out `object`.





In-Class Exercise Demo

- **DUE:**
Today 10.30PM PT
- **Submission:**
Post Github link on GAP Week 1 Day 2 Exercise